DILEEP KUMAR, Ph.D.

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Professional Summary

- **Control Systems and Electric Powertrain Engineer** with a Ph.D. and 5.8 years of experience (2.8 years in industry and 3 years in academia). Skilled in **System Engineering, Battery Systems and BMS Functions, Electric Powertrain, Vehicle Health Monitoring, Vehicle Dynamics, Control Systems, ASPICE, Embedded, and MATLAB**.
- Proficient in **BMS and Electric Powertrain System Engineering**, including requirements, architectural development, and integration, gained through working with the **System Engineering team at Mahindra & Mahindra Ltd.**.
- Proven track record of working effectively with **cross-functional teams**, contributing to the development of robust **electric powertrain software** and ensuring a smooth transition from **systems engineering to software implementation**.

Current Professional Experience — Mahindra & Mahindra Ltd.

Lead Engineer – BMS Systems Engineering **12/2023 – Present**

- Working as a **BMS Systems Function Owner** within Electric Powertrain System Engineering, while also serving as the **Quality Assurance Lead for ASPICE processes**.
- Developed vehicle energy efficiency and consumption model in Python for **vehicle health monitoring and driving range prediction**.
- Worked as a software technical delivery lead, collaborating across systems and software teams to **deliver robust electric powertrain software.**

Education

- **Ph.D., Electrical Engineering (Control & Automation)** **IIT Kanpur** (2015–2023)
  CGPA: 8.0/10
- **M.Tech, Electrical Engineering (Control Systems)** **NIT Patna** (2010–2012)
  CGPA: 8.25/10
- **B.Tech, Electronics & Communication Engineering** **SMVD University** (2006–2010)
  CGPA: 7.62/10

Technical Skills

- **System Engineering:** BMS System Requirements and Architecture, Quality, and ASPICE.
- **Control Engineering:** PID, Loop Shaping, State Space, Kalman Filter, ADRC, LQR, MPC, Disturbance Observer, Disturbance Rejection, Motor Control, and Speed & Torque Control.
- **Power Electronics:** Motor Controllers (H-Bridge, Inverter), Solar MPPT.
- **Electric Vehicles:** Vehicle/Tire Dynamics, Path Tracking Control, In-Wheel Motors, DC/BLDC motors, and Energy Efficiency and Consumption Analysis.
- **Embedded Systems:** STM32, dsPIC, ESP32, Raspberry Pi, HW-SW Co-Design, PLC.
- **Protocols & Tools:** CAN, CAN-FD, SPI, I2C, USART, CANoe, OBD, J1939.
- **Software & Simulation:** MATLAB, Simulink, Embedded C/C++, Python, Octave, LaTeX.

Past Professional Experience — NIET Jaipur

Assistant Professor – Electrical Engineering 08/2012–07/2015

- Taught Classical & Modern Control, Power Electronics, Embedded Systems, and MATLAB.
- Supervised undergraduate projects in control systems, embedded systems, motor control, and power electronics.

Research & Project Experience

1. Ph.D. Research – IIT Kanpur

- Developed solutions to improve the performance of autonomous 4WIS4WID EVs during path-tracking control.
- Analyzed and determined tank-to-wheels and well-to-wheels efficiency of battery electric vehicles, resulting in a publication in an SAE journal.
- Gained in-depth understanding of vehicle modeling, tire modeling, stability analysis, and performance evaluation.
- Designed active disturbance rejection control (ADRC) and disturbance observer-based control strategies for speed and torque tracking of DC and BLDC motors.

Control Systems, Power Electronics & Embedded Systems Related Experiences:

- Gained deeper understanding of designing and implementing motor control systems while working at the "Control Systems Laboratory" of IIT Kanpur for about 3 years.
- Built a 48V inverter for a 250W BLDC motor using the DRV8320S gate driver.
- Developed a solar MPPT system for a 100W PV panel driving a DC motor load.
- Programmed STM32, dsPIC, ESP32, Arduino, and Raspberry Pi for applications in motor control, CNC machines, and 3D printing.

2. M.Tech Research – NIT Patna

- Analysis, modeling, and simulation of a smart battery-backup system incorporating an MPPT algorithm to supply power to AC and DC loads.

3. B.Tech Project – SMVDU J&K

- Developed and demonstrated a multi-node data acquisition system using C and Assembly on 8051 microcontroller.

Scientific Contributions

- Oral Presentation: Institute Research Symposium (IRS'23), IIT Kanpur.
- Organizing Member: Indian Control Conference (ICC'18), IIT Kanpur.
- Organizer: QIP Course on "Frequency Domain Control System Design & Experiments" (2017).

Publications

1. D. Kumar, V. Jain, R. Potluri, "Energy Efficiency of Battery Electric Vehicles with In-Wheel Motors," SAE Int. Journal of Sustainable Transportation, Energy, Environment & Policy, 2022.
2. D. Kumar, R. Potluri, "On Optimal Tire Usage in the Path-Tracking Control of Four-Wheel Independent Steering Four-Wheel Independent Drive Electric Vehi- cles", SAE Int. Journal of Connected and Automated Vehicles, 2025.
3. D. Kumar, A. Kumar, "Control and Simulation of Smart Backup System for Photovoltaic System in MATLAB." International Journal of Engineering Research & Technology (IJERT) ISSN: 2278-0181, Vol. 3, 2014. IEEE ICARET, 2013.